

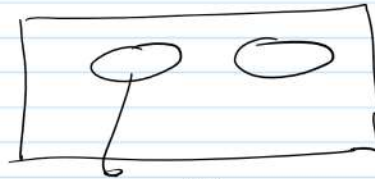
COL 774 / COL 734
Nov 14, 2025

$\mathcal{H} \subseteq \mathcal{X}$

$$\Pr \left[|\hat{\epsilon}_0(h) - \epsilon_0(h)| > \gamma \right] \leq \frac{\delta}{2e^{-2r^2m}}$$

$$z^{(u)} = 1\{h(n^{(u)}) \neq y^{(u)}\}?$$

$$\hat{\epsilon}_0(h) = \frac{1}{m} \sum_{i=1}^m z^{(u)}$$



$$\{n^{(u)}, y^{(u)}\}_{u=1}^m$$

$(n^{(u)}, y^{(u)}) \sim \text{Dist}$

$z^{(u)} \sim \text{Bernoulli}(\phi)$

$$\phi = E_{(n,y)} \sim \text{Dist} [1\{h(n) \neq y\}] = \epsilon_0(h)$$

$$\Pr(|\hat{\epsilon}_0(h) - \epsilon_0(h)| > \gamma) \leq \frac{\delta}{2e^{-2r^2m}}$$

$$\delta \geq 2e^{-2r^2m}$$

$$A_k := |\hat{\epsilon}_0(h_k) - \epsilon_0(h_k)| > \gamma$$

$h_k \in \mathcal{H} \rightarrow |h| = k$

$$\Pr(A_k) \leq \delta$$

$$\Pr(A_1 \cup A_2 \cup \dots \cup A_K) \leq \sum_{k=1}^K \Pr(A_k) \leq K \cdot \delta$$

$$\Pr(\exists h_k \in \mathcal{H}: |\hat{\epsilon}_0(h_k) - \epsilon_0(h_k)| > \gamma) \leq K \cdot \delta = \delta'$$

Uniform convergence bound

$$\delta' \geq 2Ke^{-2r^2m} \quad - (1)$$

$$\hookrightarrow \delta' \geq 2Ke^{-2r^2m} \quad - (1)$$

$$m \geq \frac{1}{2r^2} \log \frac{2K}{\delta'}$$

$$\gamma \geq \sqrt{\frac{1}{2m} \log \left(\frac{2K}{\delta'} \right)}$$

$$\hat{h} = \underset{H}{\operatorname{argmin}} \hat{L}(h)$$

↳ training error

$$h^* = \underset{H}{\operatorname{argmin}} L(h)$$

↳ generalization error

$$L(\hat{h}) \leq \hat{L}(\hat{h}) + \gamma \leq \underbrace{\hat{L}(h^*)}_{\substack{\text{with prob.} \\ \text{at least } 1-\delta'}} + \gamma \leq (L(h^*) + \gamma) + \gamma$$

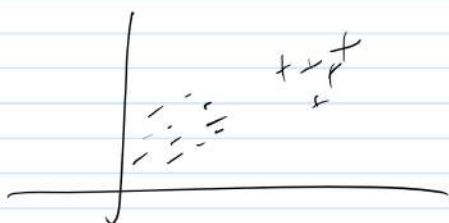
↳ h was minimum error hypothesis on training set

$$\Rightarrow L(\hat{h}) \leq L(h^*) + 2\gamma$$

with prob at least $1-\delta'$

$$\delta' \geq 2ke^{-2\gamma^2 m}$$

↳ Deal with infinite Dim Hypothesis spaces

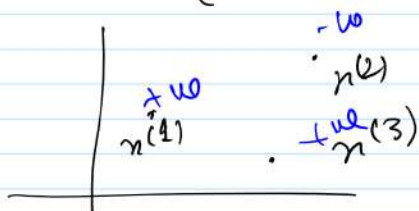


$\mathcal{H}_L^2$: set of all linear separators

$\mathcal{H}_Q^2$: set of all quadratic separators

$$\mathcal{H}_L^2 \subset \mathcal{H}_Q^2$$

$VC-D(\mathcal{H})$: Vapnik-Chervonenkis Dimension.



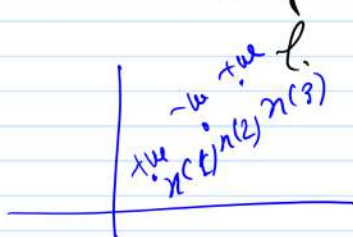
$\mathcal{X}$: instance space ($\mathbb{R}^d$)

$y \in \{0, 1\}$ $\mathcal{H}$: hypothesis class

$$S = \{x^{(1)}, \dots, x^{(n)}\} \quad x^{(i)} \in \mathcal{X}$$

We say that the set S is shattered by

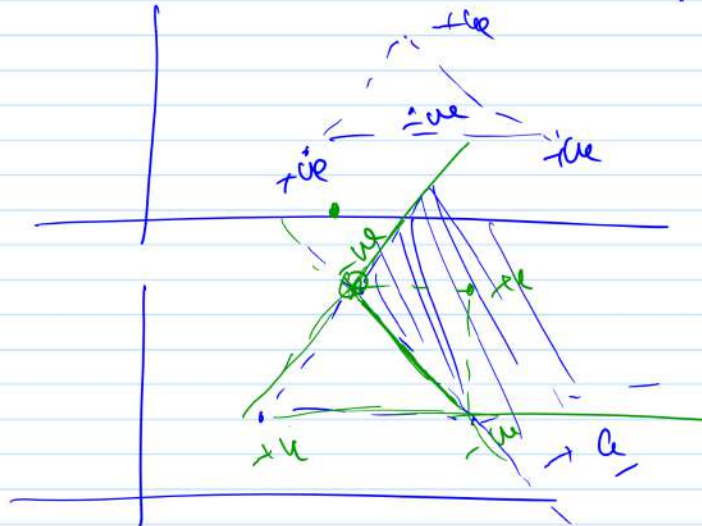
$\mathcal{H}$, if $\forall \mathbf{p} \in \{0, 1\}^n$, $\exists h \in \mathcal{H}$ s.t. $h(x^{(i)}) = p^{(i)}$



VC-D($\mathcal{H}$) :- The size of the largest set S that can be shattered by $\mathcal{H}$.

$\mathcal{H}$: Given
 $\mathcal{X}$:-

$$VC-D(\mathcal{X}_L^2) \geq 3$$



$$VC-Dim(\mathcal{X}_L^2) = 3$$

$$VC-Dim(\mathcal{X}_L^n) = \underline{n+1}$$

$$VC-Dim(H_q^2) > VC-Dim(H_L^2)$$

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